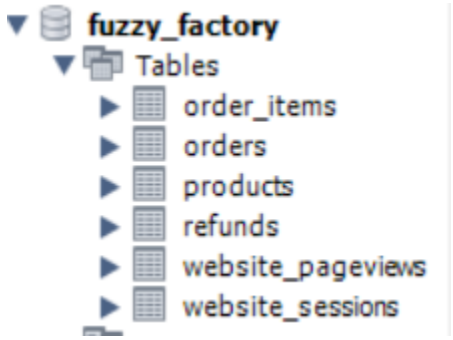
Cross-Sell Analysis

1. Project Overview

This project analyzes the scaling efficiency and cross-sell health of a multi-product operation by processing a 1.73M+ record database across 6 relational tables. The primary goal was to transform this raw transactional data into a cleaned set of 32,313 verified orders to identify high-revenue product bundles. By balancing a 171.83% YoY revenue surge against $110.04K in "revenue leakage" from refunds, this analysis isolates the most profitable cross-sell pairings to guide global marketing and quality control strategies.

2. Dataset Summary

- **Scale**: Processed **1.73M+ raw records** across **36 distinct data fields** of **total 6 tables**, covering the period from July 2012 to January 2015.



- **Cleaned Volume**: Utilized SQL to filter and aggregate the raw data into **32,313 verified orders** for the final reporting model.
- **Financial Impact**: Captured **$1.94M in Gross Revenue** and tracked **$110.04K in total refund impact**.
- **Behavioral Metrics**: Analyzed **8,000 cross-sell units**, establishing a benchmark **23.89% attachment rate**.
- **Quality Control**: Monitored **1,731 refunded units** to identify specific high-return outliers across the product catalog.

3. ETL using SQL

To ensure the accuracy of the final dashboard, a multi-stage ETL process was developed to handle the high-volume transactional data.

- **Extraction**: Queried **1.73M+ raw records** from the source database, ensuring all 36 data fields were correctly mapped for analysis.
- **Transformation**: Performed complex SQL queries to aggregate raw transactions into meaningful business metrics, establishing the foundation for longitudinal performance tracking and cross-sell evaluation.
- **Data Validation**: Performed rigorous filtering to arrive at **32,313 verified orders**, maintaining data integrity for all high-level KPIs, trend visualizations, and categorical deep-dives.
- **Load**: Structured the final datasets into optimized views for seamless integration with Power BI's DAX engine.

```sql
create view prd_performance_fact as
select o.order_id, o.created_at, o.primary_product_id, o.price_usd,
case when i.order_id is not null then 1
else 0 end as is_cross_sell,
case when r.order_id is not null then 1
else 0 end as is_refunded
from orders o
left join order_items i on o.order_id = i.order_id
and i.is_primary_item = 0
left join refunds r on o.order_id = r.order_id;
```

Final SQL transformation used to bridge 1.73M raw rows into the verified ~32K analytical dataset.

4. Key Insights & Analytical Findings (SQL)

- **Primary vs. Secondary Distribution**: To establish a baseline of market demand, I analyzed the sales split to see how many "Primary" products drive the business vs "Secondary" attachments. This confirms that while **Product 1** has the most volume as primary sales **23,861 units**, and as secondary sales **365 units**. While **Product 4** has the least as primary sales **581 units** while the highest as secondary sales **4447 units.**

product_id	pri_sales	sec_sales	total_sales
1	23861	365	24226
2	4803	993	5796
4	581	4437	5018
3	3068	1917	4985

- **Best 'Drive' for Product 4**: I identified which main product sells the most individual units to act as a "drive" for Product 4. The highest volume **"drive"** is by **Product 1** with **3,126 drive sales**.

drive_prd_id	drive_sales
1	3126
2	671
3	640

- **Attachment Rate (Prd 1 + Prd 4)**: I measured how often customers buy Product 4 when they purchase Product 1. The result was a **13.10% attachment rate**, which is below our **20% benchmark**, showing that our top-selling product is not the best at driving extra sales (cross-sell).

drive_prd_id	drive_sales	total_pri_orders	attachment_rate
1	3126	23861	13.10%

- **Highest Attachment Rate Drive**: I searched for the best "drive" product to increase Product 4 sales. **Product 3** is the most effective "drive" item, achieving a peak attachment rate of **20.86%** for making it the best candidate for bundled deals.

drive_prd_id	drive_sales	total_pri_orders	attachment_rate_prd4
3	640	3068	20.86%
2	671	4803	13.97%
1	3126	23861	13.10%

- **Refund Rate Assessment:** Since Product 3 is the primary drive for growth. I checked if certain bundles lead to more returns. And found that the combo of **Product 3 + Product 4** has a **8.13%** refund rate, which is higher than selling Product 3 alone (6.47%), suggesting that this specific bundle may introduce quality or customer expectation risks.

order_type	total_refund	total_orders	refund_rate
prd 3	157	2428	6.47%
prd3 + prd4	52	640	8.13%

- **AOV & Revenue Maximization**: I analyzed AOV across combinations to identify the most profitable bundle. This step proved that the **Product 2 + Product 4** combo creates the highest **Combo AOV ($89.98)**, outperforming the high volume Product 1 bundles ($79.98) in per-order value.

primary_product_id	total_orders	total_revenue	aov
2	4803	318109.18	66.23
1	23861	1419767.82	59.5
3	3068	180857.03	58.95
4	581	19775.72	34.04

AOV of all Products

primary_product_id	total_combo	combo_aov
2	671	89.98
1	3126	79.98
3	640	75.98

AOV of each Product + Product 4 as combo

- **Individual vs Bundle.Revenue**: I compared the total revenue of the bundle against selling items separately. The **Product 3 + Product 4** bundle generated **$48,627**, proving that bundling increases total revenue by adding, despite the **$110.04K total refund impact** across the company.

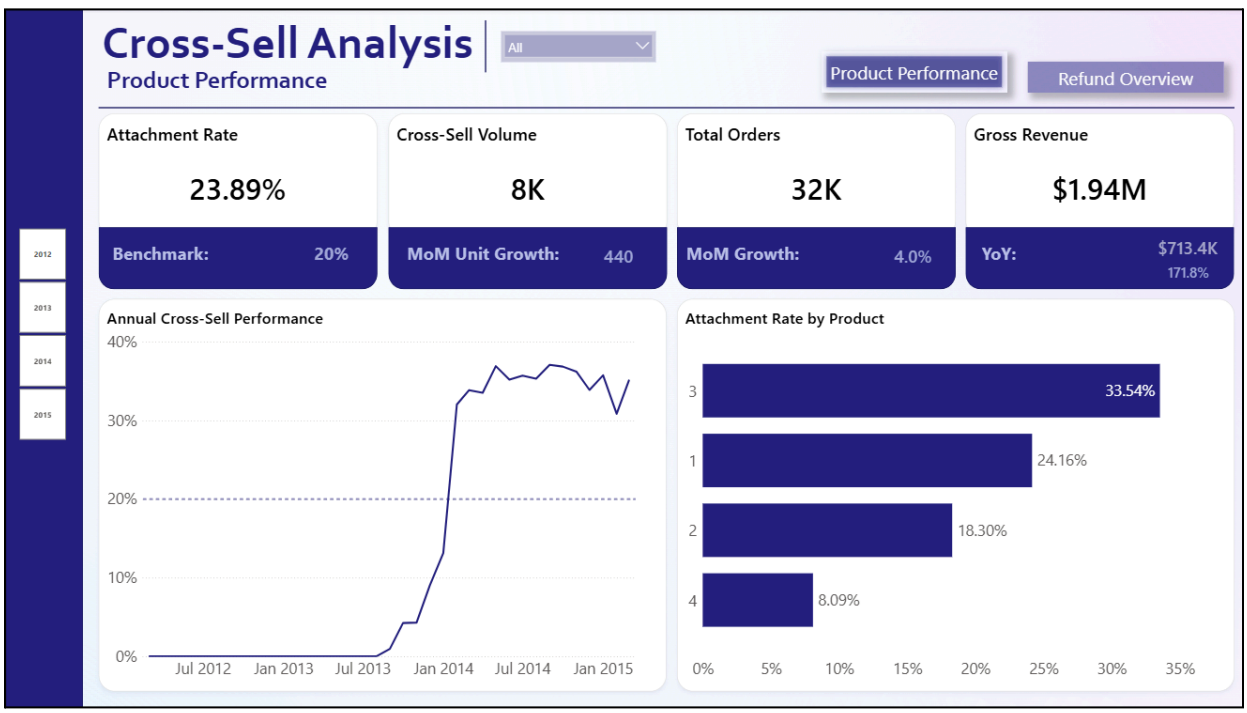
order_type	total_orders	avg_rev	total_rev
prd 3	2428	54.46	132229.83
prd 4	581	34.04	19775.72
prd3 + prd4	640	75.98	48627.2

- **Global Marketing Priority**: I compared all metrics attachment, AOV, and refund rate to find the safest product to promote globally. The data points to **Product 2** as the most stable "Global Priority" because it offers a high 13.97% attachment rate combined with the lowest **2.73% refund rate**, ensuring growth without increasing operational loss.

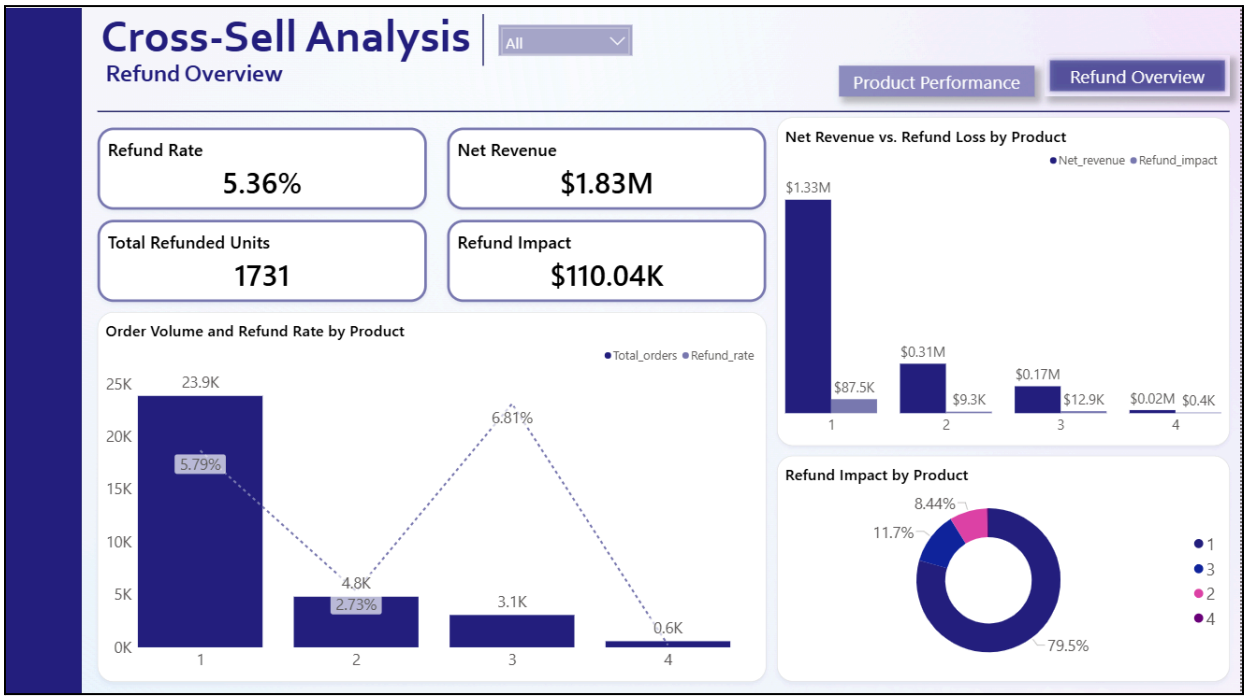
primary_product_id	total_orders	attachment_rate	aov	refund_rate
1	23861	13.10	59.51	5.79%
2	4803	13.97	66.23	2.73%
3	3068	20.86	58.95	6.81%
4	581	0.00	34.04	1.55%

5. Dashboard using PBI

Built interactive dashboard showing Product performance and Refund overview



Product Performance



Refund Overview

6. Business Recommendations

- **Scale High-Performance Bundles**: Launch targeted marketing for the **Product 3 + Product 4** combo, as it is the most effective cross-sell driver with a **21.00% attachment rate**.
- **Quality Audit for Product 3**: Investigate the **8.13% bundle refund rate** to identify if the issue stems from shipping damage or customer expectations before increasing marketing spend.
- **Prioritize Stable Growth**: For global expansion with the lowest risk, prioritize **Product 2**; it maintains a healthy **13.97% attachment rate** and the combo lowest refund risk at **2.73%**.
- **Efficiency Drive**: Improve the cross-sell strategy for **Product 1** (currently underperforming at **13.10%**), as even a 2% increase in its attachment rate would drive massive revenue due to its high sales volume.